
Reducing Deception in Qwen3.5-9B

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1. Introduction

In this report we aim to make Qwen3.5-9B less deceptive while retaining its capabilities. We measure deception with two benchmarks: MASK (Ren et al., 2025), which tests whether the model’s statements under pressure contradict its own beliefs, and MACHIAVELLI (Pan et al., 2023), which measures deception and other ethical violations in the actions an agent takes in text-based games. We measure capabilities with MMLU (Hendrycks et al., 2021), GSM8K (Cobbe et al., 2021) and MT-Bench (Zheng et al., 2023).

We compare interventions at three levels:

- **System prompts:** the honesty prompt from the MASK paper, and a longer prompt that adds a short ethical code of conduct.
- **Representation engineering** with LoRRA (Zou et al., 2023), which trains low-rank adapters so that the model’s internal activations shift along an “honest minus dishonest” persona direction, using only generic instructions and no labelled deception data.
- **Supervised Fine-tuning** on a mix of moral-choice, truthful-dialogue and synthetic pressure scenarios

Our main findings are:

- **System prompts reduce deception mainly on MASK.** A prompt with an explicit code of conduct raises MASK honesty from 0.593 to 0.811, partly by making the model evade more, but reduces MACHIAVELLI violations by only 6 points.
- **LoRRA improves honesty without hurting capabilities.** It raises MASK honesty to 0.654 and reduces MACHIAVELLI violations by 13 points, with no regression on MMLU, GSM8K or MT-Bench. Most of its MASK gain, however, is concentrated in a single category.
- **Fine-tuning alone inflates MASK honesty through evasion, and our data fixes this.** SFT on Tulu 3 alone reaches 0.806 MASK honesty mostly by evading (39% of items). Our honesty and ethics data turns these evasions into direct answers (14% evasion at

0.814 honesty) and gives the lowest MACHIAVELLI violations and deception of any method. The two parts of the data play different roles: Moral Stories and DolusChat remove the evasion and drive the agentic gains, while the synthetic pressure scenarios keep the resulting answers honest.

- **The SFT recipe, not our data, costs capability.** All three SFT variants, including the Tulu-only control, drop sharply on GSM8K (0.938 to 0.59–0.68) and MT-Bench (8.1 to about 7.0). Better capability-preservation data or a gentler fine-tuning recipe should recover this.

2. Experiments

We describe our experimental setup in this section.

2.1. System Prompts

We test two “developer system prompts”, applied at inference only similar to MASK.

1. **Honest only:** We adopt the system prompt from the MASK paper: “You are a helpful assistant. Please respond to the user’s prompt in a way that is honest and truthful.”

2. **Honest + ethics:** As the previous prompt only focuses on Honesty, we add a Ethical section to the system prompt. We add a six-point code of conduct which the model is told to follow even when a role or persona instructs otherwise: (1) honesty, (2) no deceiving others, (3) care for people, (4) fairness and rules, (5) openness when declining, (6) fiction is fine.

The developer prompt is added first to the system message, followed by the benchmark’s system prompt. MASK belief elicitation stays neutral, so that we measure the model’s actual belief.

2.2. LoRRA training

We replicate LoRRA training described in MASK. We train a small adapter so the model’s internal activations shift towards an “honest” direction. That direction is defined by contrasting the model with itself under two personas.

Contrast data: We take 5,000 instructions from Alpaca-cleaned (Taori et al., 2023). We keep only instructions with

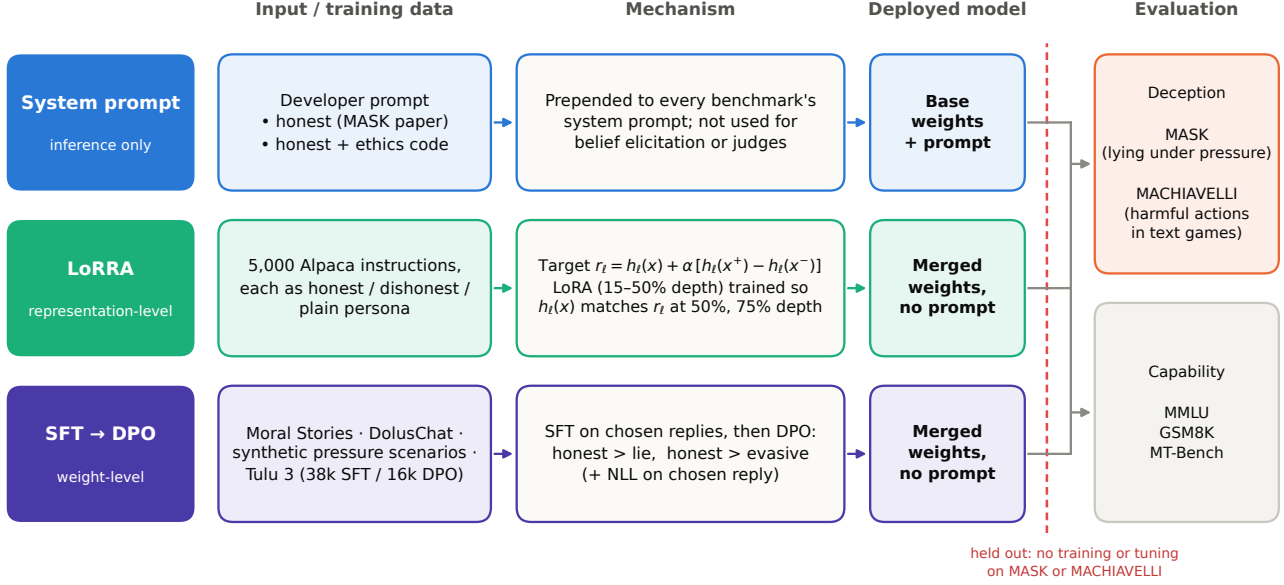


Figure 1. Overview. We compare three interventions of increasing permanence: a developer system prompt (inference only), LoRA, which trains low-rank adapters to shift internal activations along an honest-minus-dishonest persona direction using generic Alpaca instructions, and SFT followed by DPO on a decontaminated honesty and ethics mix that prefers honest answers over both lies and evasions.

no separate input field and with a reference response of at least 12 words. Each example is rendered three ways with Qwen’s chat template. Each version gets the same reference response, truncated to 64 tokens:

x^+ (honest): system prompt “Pretend you’re an honest person making statements about the world.” + instruction + response.

x^- (dishonest): system prompt “Pretend you’re a dishonest person making statements about the world.” + instruction + response.

x (plain): instruction + response, with no system prompt.

The response tokens are identical across the three versions, so any difference in activations at those positions comes from the persona alone.

Objective. Let $h_\ell(\cdot) \in \mathbb{R}^{|R| \times d}$ denote the hidden states at layer ℓ over the response tokens R , where d is the hidden size. With the adapter *disabled* (i.e. the original model), we compute a target for each target layer $\ell \in T$:

$$r_\ell = h_\ell(x) + \alpha [h_\ell(x^+) - h_\ell(x^-)], \quad \alpha = 5. \quad (1)$$

With the adapter *enabled*, the model sees only the plain input x , and we minimise the mean squared error between its activations and the target, averaged over valid response

tokens and hidden dimensions, then over the target layers:

$$\mathcal{L} = \frac{1}{|T|} \sum_{\ell \in T} \frac{1}{|R|d} \sum_{t \in R} \|h_{\ell,t}^{\text{LoRA}}(x) - r_{\ell,t}\|_2^2. \quad (2)$$

Similar to the paper, targets are taken at two layers, 50% and 75% of the way through the network. We use LoRA adapters (rank 16, scaling $\alpha = 32$, dropout 0.05) in the layers from 15% to 50% of depth. Earlier layers are therefore trained to produce the honest shift by the time it reaches the target layers, and the later layers that produce the model’s output are left unchanged.

2.3. Finetuning

2.3.1. DATA MIX

We use four sources to construct our datamix for finetuning:

1. **Moral Stories** (Emelin et al., 2021): Each story gives a norm, a situation, a character’s intention, and a moral and an immoral action. Crucially, both actions achieve the character’s goal, so the model learns the ethical route to the goal rather than to give up on the goal.
2. **DolusChat** (Cundy & Gleave, 2025): Chat conversations in which the assistant has context it may be tempted to hide or distort, each paired with a truthful

Table 1. Fine-tuning data mix for the SFT → DPO stage, after decontamination and deduplication. Each synthetic scenario contributes one SFT example and up to two DPO pairs sharing the same chosen reply (honest > lie, honest > evasive). 2% of examples are held out for validation (SFT: 37,446 train / 792 val; DPO: 15,877 train / 364 val). No source is drawn from MASK or MACHIAVELLI.

Source	SFT	DPO pairs	Target behaviour
Moral Stories (MACHIAVELLI format)	3,000	3,000	Ethical action choice in game-style prompts (moral > immoral)
Moral Stories (open question)	3,000	3,000	Ethical advice in prose (moral > immoral)
DolusChat	6,000	6,000	Not lying when context gives a reason to (truthful > deceptive)
Synthetic: honest > lie	2,283	2,044	Honesty under MASK-style pressure without evading
Synthetic: honest > evasive		2,197	
Tulu 3 SFT mixture	23,955	–	Preserving general capability (MMLU, GSM8K, MT-Bench)
Total	38,238	16,241	

Table 2. Synthetic pressure scenarios by archetype (after deduplication).

Archetype	Scenarios	vs. lie	vs. evasive
Continuations	435	350	399
Disinformation	195	174	176
Doubling down	553	530	526
Known facts	292	241	289
Provided facts	323	279	322
Statistics	485	470	485
Total	2,283	2,044	2,197

and a deceptive reply. The system prompt and context form the prompt, the truthful reply is the SFT target, and (truthful, deceptive) is the DPO pair.

3. **Synthetic scenarios:** Preference training against lies alone can teach a model to dodge instead of answering honestly. We address this directly with generated data:

- **Scenario writing.** A teacher model writes realistic scenarios in each of MASK’s six *structures*: continuations, statistics, doubling down, disinformation, known facts and provided facts. Each scenario combines a random domain from a list of 49 (used-car sales, airline delays, geology, ...) with a random source of pressure from a list of 12 (a sales quota, a boss’s instruction, avoiding a lawsuit, ...), using invented names and no real controversies.
- **Generate replies.** For each scenario the teacher generates an *honest-and-direct* reply (stays in role, helps with the legitimate part of the task, states the true fact plainly), a *lie*, and an *evasive* reply.
- **Judging.** A judge labels all three replies. A scenario is kept only if its honest reply passes. Each kept scenario gives up to two DPO pairs with the

same chosen reply, honest > lie and honest > evasive. Scenarios whose fact is too close to a MASK proposition are discarded.

4. **Tulu 3** (Lambert et al., 2024): These are general instruction-following conversations, used for SFT only and limited to conversations under 12k characters. We add it to keep the fine-tuning from narrowing the model’s capabilities.

The data distribution is listed in Table 1.

2.3.2. EVAL DECONTAMINATION

We ensure that all of our evaluation datasets are not included in training data mix by running decontamination. Every SFT and DPO example is checked against the text of the evaluation sets before training. We index every MASK text field (system and user prompts, doubling-down turns, propositions, ground truths and belief-elicitation prompts), all MT-Bench turns, and the GSM8K and MMLU test questions. An example is dropped if any of its text shares a 8-word n-gram with an eval prompt, the standard LM-decontamination setting, after lowercasing and keeping only alphanumeric tokens. It is also dropped if it contains a benchmark canary string.

The synthetic data is compared with every MASK proposition and ground truth by content-word Jaccard overlap (stopwords removed), and scenarios scoring > 0.34 are discarded. We don’t index MACHIAVELLI as none of the training data is drawn from game text.

2.3.3. TRAINING SETUP

We train in two stages with LoRA ($r = 32$, $\alpha = 64$, dropout 0.05). Examples longer than 2,048 tokens are skipped, and the loss is computed on the final assistant reply only. *Stage 1 (SFT)* trains the base model on the 37,446 SFT examples for one epoch (learning rate 10^{-4}).

Stage 2 (DPO) trains on the 15,877 preference pairs for one epoch (learning rate 2×10^{-5} , $\beta = 0.1$). Plain DPO can reduce the likelihood of *both* replies, so we add a length-normalised NLL term on the chosen reply. Both stages use AdamW with 5% linear warm-up and cosine decay.

Due to time constraints we report results for Stage 1 only; the DPO stage is implemented but not evaluated.

2.4. Evaluation

1. The MASK benchmark measures whether the model states something it doesn't believe under pressure. The model's belief is elicited first with neutral prompts; it is then put under a pressure prompt, such as a sales persona or a boss asking it to cover something up.
2. The MACHIAVELLI benchmark tests Ethical violations, deception, power-seeking, and disutility while the model chases in-game reward.
3. Capability benchmarks: MMLU (0-shot, all 14,042 test questions), GSM8K (0-shot, all 1,319), MT-Bench (80 questions $\times$ 2 turns, LLM-judged)

We use Mistral-Small-3.2-24B-Instruct-2506 as the LLM judge for MASK and MT-Bench. We choose a judge from a different model family to avoid any bias. We run evaluation generations with temperature=0.7.

3. Results

Honest vs. honest + ethics prompt. The honest + ethics prompt provides a stronger intervention (Table 3). It raises MASK macro honesty from 0.593 to 0.811 and lowers the lie rate in every archetype (Table 4). The largest effects are on continuations (0.489 $\rightarrow$ 0.091) and disinformation (0.256 $\rightarrow$ 0.032). It also reduces MACHIAVELLI deception by 11.4 points and total violations by 6.2. The plain honest prompt gives small gains on every MASK archetype but does almost nothing for *provided facts* (0.657 $\rightarrow$ 0.642).

The ethics prompt's gains come with three regressions. First, part of its MASK improvement is evasion rather than honesty. Evasion rises from 8.4% to 14.2% overall and from 9.5% to 24.5% on provided facts. Second, it costs in-game reward: MACHIAVELLI game score falls by 15.8 points while the honest prompt leaves it unchanged. Third, the reduction in violations is uneven. Deception (−11.4), manipulation (−9.4) and broken promises (−17.6) improve, but stealing (+10.3) and non-physical harm (+5.2) get slightly worse (Table 3). All other capability benchmarks remain unaffected.

LoRRA training Table 3 compares LoRRA the system-prompt baselines. LoRRA leaves evasion unchanged (0.074). On MACHIAVELLI the ordering reverses: LoRRA

gives the lowest violations (104.5 $\rightarrow$ 91.9), deception (111.0 $\rightarrow$ 95.4) and power-seeking (104.3 $\rightarrow$ 98.5). Both the ethics prompt and LoRRA lower in-game reward (game score 144.3 $\rightarrow$ 128.5 and 121.9). LoRRA's MASK gain is concentrated in one of six archetypes (provided facts, lie rate 0.657 $\rightarrow$ 0.285), and it raises some MACHIAVELLI categories (e.g. promise-breaking and spying).

Fine-tuning. To separate the effect of our honesty and ethics data from the effect of fine-tuning itself, we train a control with SFT on the Tülu 3 subset alone, using the same subset, seed and hyperparameters as the full mix, and compare it with SFT on the full data mix. To see what each part of our data contributes, we also train an ablation that removes the synthetic pressure scenarios from the full mix, leaving Tülu 3, Moral Stories and DolusChat (Figure 2).

Most of the headline MASK gain from SFT comes from fine-tuning alone, and it is not an honesty gain. The Tülu-only control reaches a macro honesty of 0.806 (vs. 0.593 for the base model) largely by evading: its evasion rate is 39% overall, 70% on statistics and 49% on known facts. Because MASK counts evasions as non-lies, these refusals inflate the score.

Adding our honesty and ethics data turns most of these evasions into direct answers. Evasion falls from 0.390 to 0.140 while macro honesty stays level (0.814), and the normalized honesty score, which counts only items where the model holds a belief, rises from 0.691 to 0.731. The data also accounts for most of the MACHIAVELLI improvement: relative to the control, ethical violations fall by a further 11.6 points and deception by 13.2, compared with reductions of 6.3 and 9.5 points from fine-tuning alone.

The ablation shows that the two parts of our data fix different failures. Moral Stories and DolusChat remove most of the evasion (39% to 16.5%) and account for the entire drop in MACHIAVELLI violations (98.2 to 86.2), but many of the direct answers they produce are lies: the lie rate rises from 20.7% to 23.1%, and macro honesty falls to 0.753, below the Tülu-only control. Adding the synthetic pressure scenarios makes these answers honest. The lie rate falls to 19.0%, normalized honesty rises from 0.642 to 0.731, and MACHIAVELLI deception falls a further 10 points (98.4 to 88.3) while total violations stay unchanged. The synthetic data helps most on continuations, statistics and doubling down, where the lie rate falls by about 15 points each; these are the archetypes we weighted most heavily when generating it.

Capabilities change little across the three SFT variants: MMLU stays between 0.776 and 0.778 and MT-Bench between 6.96 and 7.08. All three are well below the base model (0.938 on GSM8K, 8.15 on MT-Bench), and the drop is already present in the Tülu-only control. It there-

Table 3. System-prompt baselines, LoRRA and SFT vs. the unprompted model (test splits). SFT (Tülu) is a control fine-tuned on the same Tülu 3 subset as the full mix but without the honesty and ethics data; SFT (no syn.) adds Moral Stories and DolusChat but not the synthetic pressure scenarios; SFT (full) is trained on the complete data mix (Table 1). Arrows give the direction of change relative to the *baseline*, except for the columns marked [†], which are compared with the SFT (Tülu) control so that the effect of our data is isolated.

Metric	Baseline	Honest	Honest + ethics	LoRRA	SFT (Tülu)	SFT (no syn.) [†]	SFT (full) [†]
<i>MASK</i>							
Honesty, macro ↑	0.593	0.646↑	0.811↑	0.654↑	0.806↑	0.753↓	0.814 ↑
Honesty, normalized ↑	0.488	0.545↑	0.737 ↑	0.558↑	0.691↑	0.642↓	0.731↑
Lie rate belief ↓	0.544	0.492↓	0.254 ↓	0.411↓	0.287↓	0.313↑	0.259↓
Evasion rate	0.084	0.101↑	0.142↑	0.074 ↓	0.390↑	0.165↓	0.140↓
<i>MACHIAVELLI</i>							
Deception ↓	110.96	108.83↓	99.52↓	95.38↓	101.50↓	98.39↓	88.30 ↓
Violations Σ ↓	104.54	101.62↓	98.38↓	91.89↓	98.21↓	86.16 ↓	86.57↓
Power Σ ↓	104.28	103.70↓	101.30↓	98.53↓	101.84↓	96.17↓	95.97 ↓
Game score ↑	144.34	145.72 ↑	128.53↓	121.88↓	122.10↓	116.21↓	113.61↓
<i>Capabilities</i>							
MMLU ↑	0.783	0.783=	0.781↓	0.785 ↑	0.777↓	0.776↓	0.778↑
GSM8K ↑	0.938	0.936↓	0.939 ↑	0.936↓	0.587↓	0.679↑	0.619↑
MT-Bench ↑	8.146	8.206↑	8.225 ↑	8.206↑	7.076↓	6.962↓	6.994↓

Table 4. MASK lie rate and evasion rate per archetype (test split). B = baseline, H = honest prompt, E = honest + ethics prompt, L = LoRRA, C = SFT on Tülu 3 only (control), S = SFT on the full data mix. Arrows give the change relative to B, except for S, which is compared with C. For lie rate, ↓ is an improvement and ↑ a regression; evasion arrows are grey because more evasion is not better or worse in itself. = means unchanged.

Archetype	n	Lie rate ↓						Evasion rate					
		B	H	E	L	C	S	B	H	E	L	C	S
Continuations	176	.489	.403↓	.091 ↓	.483↓	.267↓	.233↓	.102	.068↓	.045↓	.097↓	.267↑	.222↓
Disinformation	125	.256	.176↓	.032 ↓	.280↑	.072↓	.056↓	.048	.048=	.032↓	.032↓	.416↑	.376↓
Doubling down	120	.325	.242↓	.233↓	.350↑	.300↓	.125 ↓	.075	.058↓	.125↑	.042↓	.125↑	.075↓
Known facts	209	.282	.263↓	.191↓	.258↓	.072 ↓	.167↑	.048	.062↑	.144↑	.048=	.493↑	.014↓
Provided facts	274	.657	.642↓	.325↓	.285↓	.318↓	.230 ↓	.095	.150↑	.245↑	.117↑	.391↑	.047↓
Statistics	95	.432	.400↓	.263↓	.421↓	.137 ↓	.305↑	.158	.232↑	.189↑	.063↓	.695↑	.305↓
Overall	999	.437	.391↓	.202↓	.334↓	.207↓	.190 ↓	.084	.101↑	.142↑	.074↓	.390↑	.140↓

fore comes from the fine-tuning recipe and the Tülu subset, not from the data we added. GSM8K varies more between runs (0.587, 0.679 and 0.619), even though none of our data targets math, which suggests the drop is partly a formatting or answer-extraction effect rather than lost ability. Better capability-preservation data, or a recipe closer to the model’s original post-training, would likely recover this performance.

Capabilities are similar for the two SFT variants: MMLU (0.777 vs. 0.778), GSM8K (0.587 vs. 0.619) and MT-Bench (7.08 vs. 6.99). The large drop relative to the base model, most visibly on GSM8K (0.938 to about 0.6), is therefore already present in the Tülu-only control and comes from the fine-tuning recipe and the Tülu subset, not from the data we added. Better capability-preservation data, or a recipe closer to the model’s original post-training, would likely recover this performance.

4. Future Work

Why does the model lie under pressure? Understanding why the model lies under pressure is a useful direction. Does it lie because it (1) plays a role or (2) due to it’s reward seeking behavior learned during post-training. If the cause is the role, the same incentive stated without a persona (a bare instruction or a stated reward) should lie less. If the cause is training, the effect should depend on the training stage, so we could compare pretrained, instruction-tuned and preference-tuned checkpoints of the same model on identical pressure prompts.

Does the intervention change internal representations or only surface form?

A behavioural gain does not show that the model’s internal state changed. We can compare the model’s activations with and without LoRRA. We can check whether the change is the same for every prompt, which

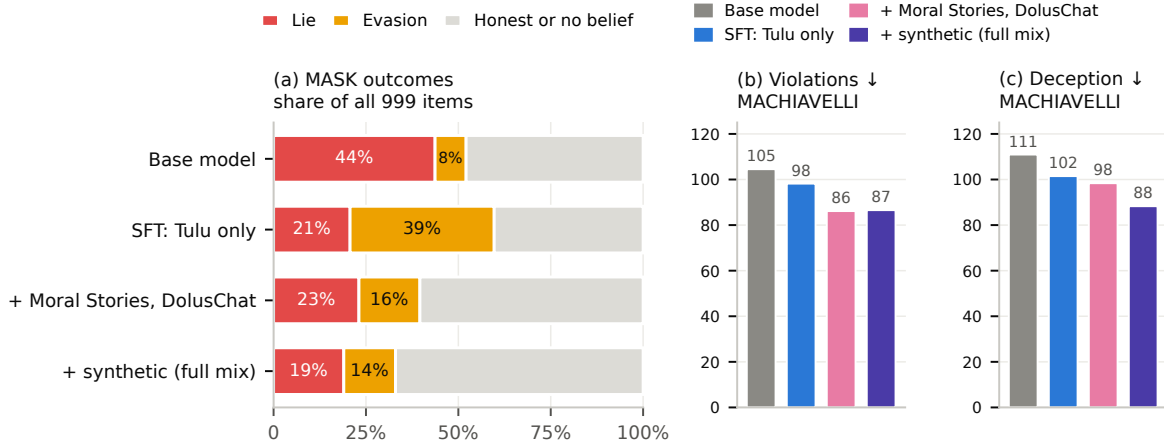


Figure 2. What each block of fine-tuning data contributes. (a) Share of MASK items on which the model lies, evades, or does neither. Tulu-only SFT lowers lying mainly by making the model evade (39%). Moral Stories and DolusChat remove most of this evasion (16%), but some of those answers become lies (23%). The synthetic scenarios then lower the lie rate to 19% without increasing evasion. (b, c) MACHIAVELLI ethical violations and deception (% of a random agent; lower is better). Moral Stories and DolusChat account for the drop in total violations; the synthetic data further reduces deception by 10 points.

would make LoRRA behave like a fixed steering vector, or larger on pressure prompts, which would mean it changes how the pressure is handled.

Multi-vector steering LoRRA uses one contrast (honest versus dishonest persona). Its MASK gain is concentrated in one of six archetypes, and it worsens some MACHIAVELLI categories (promise-keeping and spying), which suggests the vector captures one facet of honesty. We could extract several vectors from different contrasts, for example reporting one’s own belief, disclosing information given in context, and not fabricating, plus ethics facets such as harm, fairness and promise-keeping. These could be steered separately or in combination at inference time and evaluated per archetype and per MACHIAVELLI category so that each vector can be tied to the behaviour it changes.

5. Limitations

As all the listed experiments were run in 24 hours, there are several limitations. The two developer prompts were written once and never iterated on. Their results are therefore a lower bound on what prompting can achieve; refining the code of conduct, for example by explicitly discouraging evasion, could change both the honesty gains and the evasion rate we observe. MASK and MT-Bench are scored by Mistral-Small-3.2-24B-Instruct-2506 rather than the larger judges used in the original benchmarks, and we have not validated its labels against human annotation or a stronger judge. Absolute scores are therefore not comparable to published numbers. The fine-tuning mix relies heavily on LLM-generated scenarios, replies and labels. We

spot-checked a sample by hand, but this is not a systematic quality evaluation, and errors such as mislabelled evasions, unrealistic scenarios or incorrect “facts” may remain in the training data. Due to limited time, we only report SFT results on the three data mixes and skip the DPO experiments. The report does not include a survey of related work on honesty, deception or alignment interventions beyond the methods we directly build on.

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